

USER MANUAL

DEVILISHLY FRIENDLY

Commands, examples and a legion of little demons.

Legion 0.7.0 / Manual 1.4

13 NOV 2026

Your demons, clearly explained.



A guide to using Legion without wrestling with technical documentation.

Legion is firmware for the RAK3401 1W with RAK13302. It builds on Meshtastic with private tools to query, manage and recover a node.

THIS EDITION

Manual 1.4 · Legion 0.7.0 · 13 November 2026. The app reports 2.8.2.Legion0.7.0.
This edition adds MeshHealth, battery balance, sleep notices and /arm and /disarm.

Find the task you want to do

I want to…	Page
Set up LegionRF and learn the syntax	3-4
Check battery, boot and components	5-7
Send text, manage channels or restart	8-10
Find direct neighbours or trace a route	11-12
Ignore by ID or name	13
Set up antispam and use Scramble	14
Save and recover the configuration	15-16
Understand APC S1 and S2	17-19
Understand energy recovery	20-21
Update and read the daily reminder	22-23
Try a recipe or troubleshoot	24-25
Find the cheat sheets and credits	26-28

Your first query

```
/telemetry !abcdef01
```

Send it on LegionRF and replace the ID with your Legion's. You can send the query from a node running standard Meshtastic.

- Start with one query and let the demon answer.

The gateway: LegionRF.



A private channel shared by your Legion and the nodes you use to control it.

01 Create or configure an enabled SECONDARY channel named exactly LegionRF, including the correct letter case.

02 Assign your own random private 16- or 32-byte PSK. The public default key, short aliases, an empty key and all-zero keys do not enable these tools.

03 Share the channel with trusted requesting nodes. They need compatible LoRa settings to communicate.

04 Open the LegionRF chat and send the complete command as a channel message.

> **EVEN WHEN CONNECTED TO THE LEGION ITSELF**
>
> Even when your phone is connected by Bluetooth to the executing Legion, send the command in LegionRF. These tools are not invoked by direct message.

Check the gateway with a query

```
/telemetry !abcdef01
```

Use the full ID of the node that should reply: ! followed by eight hexadecimal digits. Its display name does not replace the ID.

Who can use the commands?

Access to the channel authorizes control. There is no second password. Share its key only with people who should control the Legion nodes on it.

- LegionRF is the name; use your own key. The name alone does not make it private.

One line.
One command.



Send each complete example, replacing the sample values with your own.

In this guide	Meaning
!abcdef01	The Legion that executes the command, including ignore.
!12345678	The other node: a route destination or the peer to ignore.
[page]	An optional argument. Do not type the brackets.
observe / shadow / on / off	Alternatives: choose one. Do not type them together.

Commands are lowercase. Channel names must correspond to configured channels. Do not copy headings, comments or several commands into one message.

Give the demon time

Family	Wait
Remote tools, APC and antispam	Shared 5-second cooldown after processing. Allow at least 6 seconds after a reply.
Telemetry and backup	Shared 60-second cooldown after a response attempt.
Traceroute	One active trace; at least 60 seconds between starts and up to 90 seconds for the result.

Radio scheduling can add delay. A request during cooldown may get no reply; wait the full interval before trying again.

> **THE DELIVERY ICON DOES NOT TELL THE WHOLE STORY**
>
> queued means accepted into the local queue, not received by another node. For saves and deletions, wait for the Legion’s explicit confirmation.

- If the second command gets no reply, check the cooldown first.

How is my node doing?



Battery, chip temperature, radio noise and counters in one query.

```
/telemetry !abcdef01
```

Only that Legion replies. /telemetry without an ID allows reachable Legion nodes on the channel to reply and may generate several responses.

An illustrative report

```
2.8.2.Legion0.6.6 !abcdef01 up:194400s
CPU:29.25 / max 34.50C BAT:82% 4.01 / min 3.96V
N:-116 peak:-88dBm n:600
RX:450 BAD:7 TX:32 REL:24 DROP:2
```

Field	How to read it
up	Seconds since boot.
CPU / max	Current and maximum sampled chip temperature, not ambient temperature.
BAT	Estimated percentage, physical battery voltage and observed minimum.
N / peak / n	Mean RF noise, peak RSSI and sample count.
RX / BAD	Successful and bad receptions; BAD is not exclusively CRC failures.
TX / REL / DROP	Transmissions, relays and dropped TX. REL is already included in TX.

Data lives in RAM and resets on boot. Noise is sampled on the current frequency without scanning. Meshtastic periodic telemetry and the Legion DAILY report use separate schedules.

- Physical voltage explains supply protection better than an estimated percentage.

What happened at boot?



A snapshot of startup and clues about the previous restart.

```
/boot !abcdef01
```

Field	What it tells you
reset	Hardware reset reason and full mask. Without evidence it remains unknown.
prev	Recorded previous protective stop, such as low battery or VDD. It is not a full history.
VBAT / VDD	First physical voltage samples at boot, in millivolts.
USB	USB supply detection at that sample.
NA	Unavailable data.

The detail that prevents a wrong diagnosis

The voltages in /boot belong to the start of this boot. They are not live readings or necessarily the values present when a fault occurred.

Where are the CC310 counters?

```
/components !abcdef01
```

Use /components to read them alongside radio SPI and REG1 DC/DC. The two reports keep startup evidence separate from component state.

A QUERY DOES NOT CHANGE THE NODE

/boot and /components are read-only queries. They do not save settings or reinitialize components.

- Use telemetry for current voltage; use boot for the startup snapshot.

The demons, from the inside.



CC310, radio SPI and REG1 DC/DC in one short report.

```
/components !abcdef01
```

```
!abcdef01 COMPONENTS
CC310 SHA:3 MAC:3 SW:0
SPI:ON 8MHz BULK:ON
DC/DC:ON REG1
```

Field	Meaning
CC310 SHA / MAC	Hardware SHA-256 and ACK-proof HMAC operations completed since boot.
SW	Failed accelerator attempts completed using software fallback.
SPI	Radio SPI availability and configured frequency; NA when unavailable.
BULK	Bulk radio transfers enabled in the build.
DC/DC ... REG1	Current state of the microcontroller’s REG1 regulator.

CC310 is the nRF52840 cryptographic accelerator. If an operation fails, Legion retains software fallback. Its counters reset on reboot.

ZERO DOES NOT ALWAYS MEAN A PROBLEM

MAC:0 can be normal for the traffic seen. 8MHz is the configured frequency, not a physical measurement. The report does not change SPI, DC/DC or settings.

- A counter shows use; it is not a measurement of energy savings.

Let that Legion speak.



Send text from the node you choose, even while connected to another.

```
/msg !abcdef01 SFNarrow Hello from Legion
```

Send the command in LegionRF. Legion !abcdef01 queues the text on its enabled channel named SFNarrow.

Part	Purpose
!abcdef01	Selects the transmitting node.
SFNarrow	Selects an enabled channel on that node.
Hello from Legion	The message to send there.

If the name contains spaces

```
/msg !abcdef01 "My channel" Hello everyone
```

Quote only the channel name, then add the text. Names allow up to 11 UTF-8 bytes; the whole command allows up to 199 bytes. Some characters use several bytes.

WHAT THE REPLY CONFIRMS

queued confirms local queue acceptance, not remote delivery. An invalid channel is not replaced by another, and the tool does not resend the command automatically.

- The first ID selects the speaker; the next name selects its channel.

A new channel. An available slot.



Add or remove secondary channels remotely without restarting the node.

```
/createch !abcdef01 "My channel" PSK_BASE64
/deletech !abcdef01 "My channel"
```

Replace PSK_BASE64 with the real Meshtastic PSK in standard Base64, including its padding. It is not a free-text password. Keys must decode to 1, 16 or 32 bytes; use your own random 16- or 32-byte key for privacy.

Command	Behaviour
createch	Uses the first disabled secondary slot (1-7), with a clean name and key. MQTT forwarding starts disabled.
deletech	Takes no PSK. Clears the secondary channel with that name without moving other indices.
Names	Up to 11 UTF-8 bytes; quote spaces. Existing enabled names are not overwritten and ambiguous names are rejected.

THE PRIMARY AND LEGIONRF ARE PROTECTED

deletech cannot remove them. A channel edit is rejected when Mesh Beacon is broadcasting, has a queued transmission or references that slot.

Wait for the saved confirmation. Reconnect the app if its channel list needs refreshing. Successful creation or deletion sets APC OFF and may make a backup OUTDATED; use replace only when you want to keep the new state.

- Changing channels sets APC OFF and does not refresh the manual backup.

Restart, without climbing up.



A normal restart scheduled remotely.

```
/reboot !abcdef01
```

After accepting the command, Legion schedules a normal restart 20 seconds later, using its usual save-and-restart path.

What to do

01 Send the command in LegionRF and wait for the reply.

02 Allow the 20-second delay and startup time.

03 Once it returns, query /telemetry !abcdef01. up reflects the new boot.

04 Send /boot !abcdef01 separately if you want the startup evidence.

If it says it is busy

An active settings edit, transfer, blocked recovery or insufficient supply can prevent acceptance. Wait for the operation to finish or for adequate power.

> **REMEMBER APC AND THE DAILY REMINDER**
>
> APC always starts OFF and its statistics reset. Select your preferred mode again. The daily reminder’s 24-hour clock also restarts from this boot.

A normal restart does not itself restore the rescue backup: healthy current settings remain authoritative.

- Restarting is not a factory reset.

Who can it hear directly?



Neighbours observed over radio without intermediaries.

```
/zerohop !abcdef01
```

That Legion lists nodes observed directly over RF since boot, with short names and SNR, sorted by descending SNR.

Move to another page

```
/zerohop !abcdef01 2
```

If there are more results, copy the reply’s Next: command. Allow at least six seconds after the previous report.

What you see	How to interpret it
Zero or negative SNR	Valid data. It does not mean the row is empty.
An expected node is absent	It may not have been heard directly during this boot.
partial	The capture exceeded its capacity of up to 128 rows.

AN OBSERVATION NOTEBOOK

This is not an active scan or proof that the other node hears you back. Direct reception in one direction does not guarantee both directions.

Pages use a 15-minute frozen snapshot tied to the requester and channel. Repeating page 1 reuses it, even if empty. Expiry allows a new capture; a radio change invalidates it.

- Use trace to test an outbound and return route.

Which way does the message travel?



Ask a Legion to trace another node and return both paths in LegionRF.

```
/trace !abcdef01 !12345678
```

ID	Role
!abcdef01	Legion that starts the traceroute.
!12345678	Destination node. It can run standard Meshtastic.

The trace uses the destination's recorded channel, or the primary for an unknown node.

How to read the reply

Reply	Correct interpretation
1/2 out	Outbound path.
2/2 back	Return path; it may be different.
?	An unreported hop.
0 reported relays	No relays were reported in that metadata.
NO_ROUTE	The routing system reports it as unreachable.
reachability unknown	No response arrived within 90 seconds.

ONE TRACE AT A TIME

One active trace, at least 60 seconds between starts and up to 90 seconds for a result. Other tools remain available with their cooldowns. Radio or channel changes can cancel pending results.

- A timeout does not prove that the destination is off or out of range.

This Legion ignores that node.



The command is individual: the first ID selects who acts, the second who is ignored.

```
/ignore !abcdef01 !12345678
/unignore !abcdef01 !12345678
/ignorelist !abcdef01
/ignorelist !abcdef01 2
```

Command	Result
ignore	Only !abcdef01 adds !12345678 to its effective list.
unignore	Only that executor removes the second node from its list.
ignorelist	Reads the executor’s list; the optional page follows its ID.

IT IS NO LONGER A GROUP COMMAND

The old forms without an executor are not accepted. Other Legion nodes listening on the channel do not change their lists. Existing entries are preserved.

Changes use Meshtastic’s persistent NodeDB state and include legacy entries. A reply saying in RAM; persistence pending is not yet saved. A node that misses its command does not catch up automatically. Ignoring filters packet origins. It does not stop RF transmissions or silence all unrelated traffic forwarded by that peer. Lists use a 15-minute snapshot, up to 128 rows; follow Next:. Ignore flags are outside the rescue backup.

For automatic burst filtering, see Antispam (page 14).

Ignore by name

```
/ignore name !abcdef01 "Example node"
/unignorename !abcdef01 "Example node"
/ignorelist names !abcdef01
```

Use the exact long name, including case. Rules are saved to flash.
When a node announces that name, Legion drops the announcement before replying or relaying it.
Matching IDs live in RAM. After a restart, Legion learns them again as the name is announced.

- The first ID selects who acts. Each Legion keeps its own rules.

A demon against spam.



Guard watches public text received over LoRa on the primary channel.

Commands on LegionRF

Command	Result
`/guard status !abcdef01`	Read the mode, counters and quarantines.
`/guard observe !abcdef01`	Simulate quarantines. This is the boot mode.
`/guard on !abcdef01`	Enable automatic blocking on that Legion.
`/guard off !abcdef01`	Stop detection and clear temporary evidence.
`/guard clear !abcdef01 !12345678`	Clear that node's evidence and quarantine.
`/spamlist !abcdef01`	Open the quarantine list at page 1.
`/spamlist !abcdef01 2`	Read page 2 from the same snapshot.

The first ID is the Legion that runs the command. With clear, the second ID is the affected node. Allow at least 6 seconds after a reply before sending the next command.

When the filter acts

Each sender starts with 8 credits and recovers 1 every 5 seconds. In ON, the filter drops messages received without credit. Four excess messages or six identical texts with distinct IDs within 20 seconds trigger a 10-minute quarantine.

The filter covers primary-channel public text, including compressed text. Copies of the same packet ID do not count. Quarantines are temporary and live in RAM. The manual /ignore list on page 13 is separate.

DAILY AND COMBINED CONTROL

SPAM:OBSERVE simQ:1 is simulated. SPAM:ON Q:1 is active. /arm !abcdef01 turns APC and antispam on together. /disarm !abcdef01 turns both off. See page 17 for the replies.

Scramble · emergency silence

`/scramble !abcdef01`

Normal traffic stops on receipt. The entry notice has up to 4 s to transmit.

`Entering hibernation for Scramble.`

LoRa, Bluetooth and USB data stop. After 60 min, the node restarts.

`Hibernation ended. Scramble complete.`

Power loss or a physical reset can end Scramble early.



- See Antispam on GitHub for the full filtering rules.

Save a point to return to.



Create the backup explicitly. Installing, restarting or changing settings does not refresh it.

Create, inspect and refresh

```
/backup !abcdef01
/backup status !abcdef01
/backup replace !abcdef01
```

backup creates a point if absent; status only queries; replace explicitly updates it with current settings and avoids another write if unchanged.

THE CONFIRMATION TO WAIT FOR

saved and verified; recovery armed confirms the save, verification and armed recovery. Telemetry and backup share a 60-second cooldown.

State	Meaning
up to date; recovery armed	Saved blocks match current settings.
OUTDATED: owner,channels	Those blocks differ; the previous rescue point remains usable.
freshness unavailable	A reliable comparison is currently unavailable.
no rescue point	There is no active rescue point.
invalid or incompatible	The record cannot be used.

It can identify owner, config, modules, channels and position without exposing keys or private settings. Dynamic GPS does not make it stale. Creating or deleting channels may change freshness; the daily reminder uses the same comparison without saving another backup.

- After a change you want to keep, query status and decide whether to use replace.

Let it rise again with its identity.



Restoration acts on confirmed loss or corruption of essential preferences.

What comes back, and what does not

It saves names, private identity and coherent public key/ID, channel keys, device and radio settings, modules and fixed position. It excludes conversations, peer databases, dynamic GPS, counters, queues and BLE bonds. Ham/licensed and “cannot be messaged” flags use their normal defaults.
Healthy current settings remain authoritative even if they differ from the backup. A restart, name change or ordinary update does not trigger restoration.

If you want blank settings

```
/backup delete !abcdef01
```

BEFORE AN INTENTIONAL FACTORY RESET

A factory reset restores an armed rescue point. For blank settings, delete the point first, wait for deleted; recovery disabled, then request the reset.

The return notice

Ashes to ashes, dust to dust, the node [node_name] has risen again
After successful recovery, it is queued on the restored LegionRF channel. A failed notice does not undo recovery. Incomplete restoration keeps the radio silent and has a bounded retry limit.
The flash record has integrity protection, not encrypted secret storage. Deletion disables recovery without forensic erasure. A full-chip erase or incompatible firmware can destroy it.

- The backup keeps one chosen configuration; also keep an external export.

Care for the battery, with different modes.



A single setting combines both APC stages and keeps reception active.

S1 budgets airtime for optional relays. S2 extends that policy by choosing temporary TX power for each eligible relay that goes on air. The configured power remains the reference, and the node's own traffic uses it.

Mode	Behaviour
OBSERVE	Collects voltage and TX statistics. Relays use nominal power.
SHADOW	Models the S1 budget and S2 reductions while relaying at nominal power.
ON	Applies S1 and S2 after warmup and the required time to enter each state.
OFF	Restores normal forwarding and nominal power.

```
/apc observe !abcdef01
/apc shadow !abcdef01
/apc on !abcdef01
/apc off !abcdef01
```

Use the CLIENT role and send one command per message on LegionRF. APC boots in OFF. Its mode and counters live in RAM.

MODES CHECKED ON HARDWARE

OBSERVE, SHADOW and ON were tested. Channel messages and DMs kept working in CONSERVE and RESERVE during those tests.

Repeating the current mode keeps the session. Switching active modes keeps the counters and restarts warmup. OFF retains the stopped session's results. The next activation starts a new session.

SWITCH APC AND ANTISPAM TOGETHER

Send either command on LegionRF. APC warmup still applies. Both modes are kept in RAM.

```
/arm !abcdef01     → !abcdef01: APC ON (S1+S2). SPAM ON.
/disarm !abcdef01  → !abcdef01: APC OFF. SPAM OFF.
```

- OBSERVE watches; SHADOW models; ON applies; OFF restores normal admission.

Not every packet is subject to limits.



Battery state guides both the relay budget and transmit power.

When the state changes

Transition	Filtered VBAT	Continuous time
NORMAL → CONSERVE	≤ 3.80 V	15 min
NORMAL / CONSERVE → RESERVE	≤ 3.65 V	5 min
RESERVE → CONSERVE	≥ 3.75 V	15 min
CONSERVE → NORMAL	≥ 3.90 V	15 min

APC first needs 5 minutes of fresh, valid samples. The timers above then begin. In NORMAL, with detected USB or with invalid or stale samples, forwarding follows normal admission at nominal power. Solar charging can still show USB:0.

How much is relayed and at what power

S1 budgets and S2 targets with 27 dBm configured. S2 acts when a packet goes on air.

Adjustment	CONSERVE	RESERVE
S1 · Airtime budget for optional relays	600 ms/min	120 ms/min
S2 · Position, telemetry and unchanged NodeInfo after TX	24 dBm	21 dBm
S2 · Relayed text and opaque PKI direct messages	24 dBm	24 dBm
S2 · Own traffic, LegionRF and control	27 dBm	27 dBm

S1 covers only decoded third-party position or telemetry broadcasts received over LoRa, with hops left, normal or lower priority, no ACK or response request and outside LegionRF. Unknown or oversized airtime estimates follow normal admission. Credit builds up to 2,000 ms and empties on a state change.

S2 lowers power by 3 or 6 dB as shown, with a 21 dBm floor. At 21 dBm or below it uses the nominal setting. First or changed NodeInfo and packets the node has already transmitted with the same ID use nominal power.

- These voltage thresholds are not calibrated battery percentages.

The report, page by page.



Seven pages from one snapshot, with S2 diagnostics on page seven.

```
/apc status !abcdef01
/apc status !abcdef01 2
/apc status !abcdef01 7
```

Start with page 1 to create a snapshot valid for 15 minutes. Then ask for any page from 2 to 7. Requesting page 1 again replaces it. Allow at least 6 seconds after a reply before sending the next command.

Page	Contents
1	Mode and state, inhibition, raw and filtered VBAT, and decisions.
2	Duration, samples, USB, battery statistics and current and minimum VDD.
3	Starts, completions, aborts and estimated airtime for own TX and relays.
4	IRQ timing, repeated IDs, start failures, open TX, gaps and configuration changes.
5	Radio, configured power and compensated chip-drive setting.
6	S1 budgets, protected admissions, state changes and refused airtime.
7	S2 RF targets, faults and counts of nominal, reduced and simulated TX.

Three S1 counters

eligible counts eligible attempts, would the simulated refusals and skip the refusals applied in ON. In OBSERVE they can stay at zero while packets are relayed.

RF TARGETS AND ESTIMATES
The dBm values are firmware targets. RF output, range and battery savings have yet to be measured. evict counts replacements in the 32-ID tracker. A completed TX records the end of local transmission.

Relevant settings or channel changes, restart, shutdown, a transfer or recovery disarm APC. Reactivate it explicitly. The daily reminder only summarises its state.

- Session and report IDs identify pages belonging to the same snapshot.

Two voltages. Two stories.



Energy recovery is automatic and independent of APC and the backup.

Reading	What it measures	When
VBAT	Physical voltage through the RAK19007 divider.	Every 5 s, at boot and before TX.
VDD	Internal nRF52840 supply.	Every 1 s and during boot, TX and flash checks.
USB / VBUS	USB detected at the node.	At decisions that use it.

The policy targets RAK3401 + RAK13302 with battery and RAK19007: one Li-Ion/LiPo cell or parallel cells. Four parallel 18650 cells still have single-cell voltage.

The physical battery reading

VBAT averages fifteen 12-bit ADC readings with a 3.0 V reference. R3 = 1 MΩ and R4 = 1.5 MΩ: the ADC sees about 60% of VBAT. VDD is measured separately using the internal 3.6 V range.

Physical voltage does not depend on the app's ADC multiplier. Supply protection and the app's estimated percentage can therefore disagree.

> **A SOLAR PANEL DOES NOT HAVE TO LOOK LIKE USB**
>
> The charger may charge the battery while the node shows USB:0. These readings do not measure solar current, capacity or stored energy.

Use /telemetry for current/minimum VBAT, /boot for first samples, and /apc status for statistics from an active session.

- VBAT describes the battery; VDD describes the microcontroller's supply.

When power runs low, the demon rests.



Supply protection avoids booting, transmitting or saving with insufficient power.

Protection	Threshold and behaviour
Boot	VBAT ≥ 3.60 V and VDD ≥ 3.00 V for four checks 250 ms apart; repeated after enabling IO2.
Low battery	Eight consecutive checks below 3.50 V, every 20 s, lead to energy sleep.
Low VDD	On battery: unsafe below 2.90 V, safe again from 3.00 V. Six unsafe checks, normally 1 s apart, lead to sleep.
TX reserve	VBAT ≥ 3.50 V and VDD ≥ 3.00 V; checked before the attempt and again before launching TX.
Saving	Battery reserve ≥ 3.50 V and VDD guard: unsafe below 2.90 V, safe again from 3.00 V.

The first to eighth low reading spans nominally 140 s; actual response depends on the crossing time and scheduling. TX and flash do not wait for all eight checks.

How it returns

Sleep turns off radio and LEDs, cuts IO2 and radio power, and stops reception and relays. LPCOMP can wake the MCU as battery voltage rises: nominally about 3.78 V at 3.30 V VDD, subject to circuit, hysteresis and tolerances.

If that wake source cannot be armed, it waits with radio off: four healthy checks every 30 s, VBAT ≥ 3.80 V and VDD ≥ 3.00 V, before resetting. Healthy USB-powered VDD bypasses the battery count.

TWO PRACTICAL DETAILS

Returning goes through boot checks again. An intentional shutdown from the app or CLI does not arm this battery wake source.

LOW SUPPLY NOTICES ON LEGIONRF

On battery, three readings 5 s apart at VBAT ≤ 3.65 V or VDD ≤ 3.10 V can trigger a prewarning while TX is possible. After an energy wake, the second message reports the return to service.

```
Low supply voltage. Hibernation possible.
Hibernation ended. Supply voltage recovered.
```

- USB bypasses applicable battery thresholds; VDD and flash protection still matter.

What it does itself. And how to update.



Not every improvement needs a command in the chat.

Improvement	What it adds
8 MHz SPI	SX1262 buffer transfers in blocks through nRF52 SPI/EasyDMA.
REG1 DC/DC	Efficient MCU regulator after SoftDevice initialization.
CC310	Hashes and HMAC reported by /components, with software fallback.
Persistence	Skips identical saves and verifies length/CRC32 before replacing critical files.

These measures do not establish a savings figure. Current consumption and battery-life gains have not been measured. The large NodeDB retains upstream’s non-atomic replacement.

Update a Legion application

01 Download the release for your kit from the repository’s Releases.

02 Use UF2 over USB, or the ZIP with nRF Connect for OTA.

03 An ordinary application update preserves preferences: no erase or factory reset is required.

04 Check the version with /telemetry and rescue with /backup status, respecting their cooldowns.

OTA AND RESCUE

The 0.6.5 candidate was installed successfully via OTA. Earlier releases remain available for rollback. Compatible application updates preserve rescue data; a full-chip erase may remove it.

AUTOMATIC CPU THERMAL HIBERNATION

The chip is sampled every 30 s. Three minutes at 70 °C or above trigger hibernation. Two valid readings in a row at 60 °C or below trigger a restart. If Scramble overlaps, its timer also applies.

```
Entering hibernation due to high CPU temperature.
Hibernation ended. CPU cooled down.
```

The first LegionRF notice has up to 4 s to transmit. The second follows the restart. /boot reports prev:cpu-thermal.

- After an update, APC starts OFF and the backup is not refreshed automatically.

One reminder a day. No request needed.



LegionRF receives one report every 24 hours from boot.

```
!abcdef01 DAILY
APC:ON NORMAL
SPAM:ON Q:0
BACKUP:UP TO DATE
BAT:78% (-5pp) 4.01V (-60mV)
MeshHealth 60/100
Heard:258 Direct:18 2+ hops:59
Relayed:120 Sources:54
DM targets:5 Confirmed:3
```

HOW TO READ THE REPORT

APC / SPAM	APC mode and state. simQ counts simulated quarantines and Q counts active ones.
BACKUP	Rescue point status. OUTDATED means saved settings differ from current settings.
BAT	Estimated charge and measured voltage. Changes since the last cut use points and mV. 24h:NA means a reference is missing.
MeshHealth	Score from 0 to 100 against up to seven comparable cuts. A 100 meets the node's reference. NA means no reference. PARTIAL marks incomplete data.
Heard / Direct / 2+ hops	LoRa origins heard, those heard directly, and those whose shortest known path used at least two relays. MQTT origins are excluded.
Relayed / Sources	Completed local LoRa relays and the number of distinct origins they carried.
DM targets / Confirmed	Direct text destinations and those that confirmed with an explicit ACK in the same cut.

WHEN THE DAILY USES TWO MESSAGES

If it exceeds 199 bytes, DAILY 1/2 has APC, SPAM, BACKUP and BAT. DAILY 2/2 has MeshHealth. Both use the same 24-hour cut. A busy node can retry pending parts once a minute for 15 minutes.

- The clock starts at the latest boot; restarting begins the count again.

Four tasks. Straight to the point.



Replace the IDs and send each line separately in LegionRF.

1 · Inspect a remote node

```
/telemetry !abcdef01
/boot !abcdef01
/components !abcdef01
```

Read battery and counters, then boot evidence and components. Allow time between messages and respect cooldowns.

2 · Save a change you want to keep

```
/backup status !abcdef01
/backup replace !abcdef01
```

Query, wait at least 60 s and replace if you want to keep the settings. Wait for saved and verified; recovery armed. If there was no point, create one with /backup !abcdef01.

3 · Check whether the link is direct

```
/zerohop !abcdef01
/trace !abcdef01 !12345678
```

The first shows direct observations; the second tests a route to the other node. Allow at least 6 s after the first reply and up to 90 s for the trace.

4 · Model S1 and S2

```
/apc shadow !abcdef01
/apc status !abcdef01
/apc status !abcdef01 7
```

Give SHADOW time to collect activity and model reductions. Then open a snapshot with status and read page 7 for S2. Allow at least 6 seconds after each reply. Finish with /apc off !abcdef01.

- A recipe is not a block to paste at once: each line is a separate message.

Before summoning more demons…



Common questions have a specific check.

What happens	What to check
No reply.	Enabled SECONDARY, exact LegionRF name, valid private PSK, compatible radio, lowercase command and full ID.
I sent a DM.	Send it as a LegionRF channel message.
One reply, then silence.	60 s for telemetry/backup; at least 6 s after remote tools/APC replies.
queued, but not seen.	Only the local queue was confirmed. Check channel, coverage and traffic.
zerohop does not change.	It reuses its snapshot for 15 minutes; it is not a live scan.
trace says unknown.	No reply in 90 s; it does not prove the destination is unreachable.
Backup OUTDATED.	The named blocks differ; use replace only to keep the change.
My settings return after factory reset.	Rescue was armed. For blank settings, delete it first and wait for confirmation.
APC returns to OFF.	Expected after boot or relevant channel/settings changes; reactivate explicitly.
OBSERVE has zero would/skip.	That mode does not make limiting decisions; normal forwarding can continue.
Another Legion’s ignore is unchanged.	Only the first ID acts: address each executor separately.
Solar charges but USB is 0.	External solar charging may not trigger detected VBUS.

Record the exact version, command and reply. Add telemetry, boot, components and the APC report if it was active.

- The exact reply is more useful than repeating commands rapidly.

The commands you will use most.



Send all these examples as messages in the private LegionRF channel.

Node state

```
/telemetry !abcdef01
/boot !abcdef01
/components !abcdef01
```

Current telemetry, boot evidence and components. /telemetry without an ID allows several Legion nodes to reply.

Messages, channels and restart

```
/msg !abcdef01 SFNarrow Hello from Legion
/createch !abcdef01 "My channel" PSK_BASE64
/deletech !abcdef01 "My channel"
/reboot !abcdef01
```

The PSK is Base64; deletech takes no key. Channel edits set APC OFF. A restart is scheduled after 20 s.

Direct neighbours and routes

```
/zerohop !abcdef01
/zerohop !abcdef01 2
/trace !abcdef01 !12345678
```

Ignore manually

```
/ignore !abcdef01 !12345678
/unignore !abcdef01 !12345678
/ignorelist !abcdef01
/ignorelist !abcdef01 2
```

Antispam

```
/guard status !abcdef01
/guard observe !abcdef01
/guard on !abcdef01
/guard off !abcdef01
/guard clear !abcdef01 !12345678
/spamlist !abcdef01
/spamlist !abcdef01 2
```

ALWAYS INCLUDE THE EXECUTOR

The first ID is the Legion that acts. With ignore, unignore and guard clear, the second ID is the affected node. With ignorelist and spamlist, the optional page follows the ID.

Ignore by name

```
/ignore name !abcdef01 "Example node"
```

Scramble

```
/scramble !abcdef01
```

- Replace IDs: ! followed by eight hexadecimal digits.

Save, observe and return.



A quick reference for configuration and energy tools.

Rescue point

```
/backup !abcdef01
/backup status !abcdef01
/backup replace !abcdef01
/backup delete !abcdef01
```

Form	Quick meaning
backup	Create if absent.
status	Query validity and freshness without updating.
replace	Explicitly replace with current settings.
delete	Disable persistent rescue; wait for confirmation.

APC S1 and S2 · CLIENT role

```
/apc observe !abcdef01
/apc shadow !abcdef01
/apc on !abcdef01
/apc off !abcdef01
/apc status !abcdef01
/apc status !abcdef01 7
```

ON enables both stages. SHADOW models them at nominal power. Run status without a page number to open the snapshot, then request page 7 to read relay power. APC boots in OFF.

COOLDOWNS AT A GLANCE
Telemetry and backup share 60 seconds. Remote tools, APC and antispam share a 5-second cooldown. Allow at least 6 seconds after a reply. Trace requires 60 seconds between starts and can take up to 90 seconds.

DAILY reports APC, antispam, backup, battery change and MeshHealth every 24 hours from boot.

- For a blank factory reset: backup delete, confirmation, then reset.

Many demons.
One shared manual.



A guide to accompany Legion releases, in Spanish and English.

Edition details

Item	Reference
Manual	Legion friendly user manual · English · edition 1.4.
Firmware	Legion 0.7.0 · 13 November 2026.
Identification	2.8.2.Legion0.7.0.
Meshtastic base	2.8.2 develop · d5f1619.
Hardware	RAK3401 1W + RAK13302; energy policy for battery and RAK19007.
Editorial review	DAILY, combined controls and thermal hibernation.

Further reading

Repository and downloads: https://github.com/ethan1980/RAK3401-Legion
Commands: COMMANDS.md · Rescue: RECOVERY.md
Energy: ENERGY-RECOVERY.md · APC: APC.md · Antispam

Project and acknowledgements

Legion is Ethan (Javier)’s project, based on Meshtastic and its community; it retains the firmware’s GPL-3.0 licence. It credits NavaTastic for resilience inspiration and Nexus/Jentastic for the command system.
This edition keeps the official logo and original black-and-red demon illustrations. The Spanish and English versions follow the same organization and are linked from the repository.

- Unleash your demons.